

chondrosarcoma-mets-acetabular-fx-id h-unknown-y34r

Preclinical recommendations — forward-looking horizon scan

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Preclinical recommendations

A forward-looking horizon scan of candidate drugs, compounds, and treatment strategies that are earlier than clinical development, or not yet developed, but plausibly target this patient's stated features. These are research directions, not enrollable options or treatment recommendations.

1. BCL-xL-selective inhibitors (A-1155463, A-1331852, WEHI-539) with doxorubicin or cisplatin (combination)

Target: High-grade chondrosarcoma with dedifferentiated features: the apoptotic block that limits chemotherapy

Stage: In vivo (animal) **Evidence strength:** Moderate

Chondrosarcoma-specific in-vivo activity and a named dependency, but the anchor paper's chemotherapy synergy held only in culture, and only in three of eight lines.

Chemotherapy is the one systemic lever this histology actually offers, so the most useful preclinical question is why chondrosarcoma cells absorb an anthracycline dose and survive it. de Jong's dissection gives that resistance a name. Across eight chondrosarcoma lines, selective BCL-xL blockade killed cells and synergised with doxorubicin or cisplatin in three of them, while a BCL-2-selective compound did nothing at all. BCL-xL protein also rose with histological grade across 137 tumours, which points the finding toward her end of the disease rather than away from it. The weak point sits exactly where the clinical hypothesis lives: in the orthotopic rat, BCL-xL inhibition slowed growth on its own but failed to sensitise to doxorubicin. Every line and every tumour on that array was conventional chondrosarcoma, so the dedifferentiated component has never been tested, and there is no assay to pick the three-in-eight responders in advance. It sits first because it is the only idea in this scan with in-vivo chondrosarcoma activity attached to a named, measurable dependency, not because anyone is close to dosing a person with it.

Mechanism. BCL-xL holds BAX and BAK in check and keeps chondrosarcoma cells above the threshold where anthracycline damage would trigger mitochondrial permeabilisation. A BH3 mimetic that binds the BCL-xL groove selectively displaces those partners, so a chemotherapy dose the cell was already absorbing can convert into cell death.

Translatability. Moderate: eight chondrosarcoma lines and an orthotopic rat model, though the in-vitro chemo synergy did not reproduce in vivo and no dedifferentiated model was tested.

Developability. Far. A-1155463, A-1331852 and WEHI-539 are tool compounds with no IND. A clinical-grade route would run through platelet-sparing BCL-xL degraders.

Counter-productive MoA. Moderate (On-target thrombocytopenia compounds anthracycline myelosuppression, capping the chemotherapy dose the combination is meant to rescue.)

Key risks.

- On-target thrombocytopenia is the reason no BCL-xL-selective small molecule has been developed as a systemic agent.
- The in-vitro chemotherapy synergy did not carry into the rat model, which is the step the whole clinical idea depends on.

- Only three of eight lines synergised, and nothing exists to identify a responder prospectively.
- The cell lines and the 137-tumour array are conventional chondrosarcoma, so the dedifferentiated component that defines her tumour is untested.

Open questions.

- Does the dedifferentiated component carry the same BCL-xL dependence as the conventional lines that were tested?
- Can BH3 profiling on clinical tissue separate the responders from the five lines that did nothing?

Key references. pmid:30242253; pmid:33266275

2. DNMT1-directed hypomethylating therapy (decitabine, guadecitabine) in H3K27me3-deficient dedifferentiated chondrosarcoma (repurposed drug)

Target: Dedifferentiated component with PRC2 inactivation: a target inside the transition itself **Stage:** In vivo (animal) **Evidence strength:** Emerging

One cheap stain would settle whether she is even a candidate, but every therapeutic result behind the idea was generated in a different sarcoma.

Most of the chondrosarcoma literature is generated in conventional disease and then pointed at the dedifferentiated variant. This is one of the few ideas aimed at the dedifferentiated component directly. Makise found H3K27me3 loss in six of nineteen dedifferentiated chondrosarcomas, confined to the dedifferentiated areas while the cartilaginous component kept the mark, with SUZ12 or EED alterations tracking the same distribution. Patel then showed in malignant peripheral nerve sheath tumour that precisely this state creates a DNMT1 dependency, with antitumour response in vivo. What makes it worth writing down is how cheap the question is to ask: a single immunohistochemical stain, which the expert bone-sarcoma pathology review this case already needs could carry at no extra tissue cost, and hypomethylating agents that are approved and cheap. What keeps it from ranking higher is that not one chondrosarcoma model has ever been treated with a hypomethylating agent, the therapeutic evidence is entirely a different sarcoma, and H3K27me3 loss skewed toward ribs and the upper body rather than the pelvis, which lowers her prior before any staining is done. Hypomethylating agents also have a long record of disappointing in solid tumours, and nothing here should hold up chemotherapy.

Mechanism. Losing PRC2 strips H3K27me3 repression off retrotransposons and leaves DNA methylation as the only remaining brake. Inhibit DNMT1 and that brake comes off too: derepressed retroelements form double-stranded RNA, the PKR-dependent viral-mimicry response fires, and the cell dies. Cells with intact PRC2 keep the histone-mark safeguard and ride out the same exposure.

Translatability. Moderate on the biomarker, low on the drug: the H3K27me3 loss is human dedifferentiated chondrosarcoma, the DNMT1 therapeutic data are entirely nerve sheath tumour.

Developability. Close on molecule, far on evidence: decitabine and azacitidine are approved and cheap, and the selection assay is one immunohistochemical stain.

Counter-productive MoA. Moderate (Hypomethylating myelosuppression stacks with the anthracycline chemotherapy this histology warrants, and could cost her the better-supported treatment.)

Key risks.

- The therapeutic data come entirely from malignant peripheral nerve sheath tumour; no chondrosarcoma model has been treated.
- H3K27me3 loss appeared in about a third of cases and skewed to ribs and the upper body, so a pelvic lesion starts with a lower prior.
- Loss of the histone mark on staining is not the same thing as a PRC2 inactivating mutation, and the two can come apart.
- Myelosuppression from a hypomethylating agent stacks badly with the anthracycline-based chemotherapy this histology warrants.
- Hypomethylating agents have disappointed repeatedly in solid tumours.

Open questions.

- Does her dedifferentiated component actually lose H3K27me3, given the site skew away from the pelvis?
- Would a hypomethylating agent do anything in a chondrosarcoma model, which nobody has yet tried?

Key references. pmid:35789380; pmid:30291346; pmid:30898839

3. NAMPT inhibitors (FK866/APO866, KPT-9274, OT-82) exploiting NAPRT promoter hypermethylation (small molecule)

Target: IDH1/IDH2 mutation status: the founder event's methylation debt rather than the enzyme **Stage:** In vitro
Evidence strength: Emerging

Survives the case's central uncertainty, since sensitivity tracked NAPRT methylation rather than IDH status, but the drug class has a discouraging clinical record and no chondrosarcoma animal data.

The case record already argues that IDH dependence looks outgrown in dedifferentiated tissue. Removing D-2-HG cut it by more than 90% with no phenotypic change out to 20 passages, and the one published IDH-inhibitor figure in this subtype is a median progression-free survival of 1.5 months. That is precisely why this row targets the methylation debt the founder event left behind rather than the enzyme. Peterse found low-nanomolar sensitivity in nearly half of eleven chondrosarcoma lines, and sensitivity tracked NAPRT expression and promoter methylation with explicitly no correlation to IDH1/2 status. The vulnerability is a consequence of the hypermethylation phenotype, which persists whether or not the dedifferentiated clone still needs the mutant enzyme, and it survives the scenario where her IDH testing comes back wild-type. One detail runs in her favour: NAPRT promoter methylation was higher in high-grade than low-grade tumours. Set against that, the chondrosarcoma case is cell-line work from a single group with no animal data at all, and the drug class has a genuinely bad clinical history. FK866 and GMX1778 stalled in phase 1 on thrombocytopenia and retinal toxicity without an efficacy signal. There is also no NAPRT assay that works on clinical material, so the selection biomarker this idea depends on cannot currently be measured in a patient.

Mechanism. A cell can build NAD⁺ from nicotinamide through NAMPT or salvage it from nicotinic acid through NAPRT. Methylate the NAPRT promoter and the second route closes, leaving the cell wholly dependent on NAMPT. Block NAMPT and the NAD⁺ pool collapses, taking glycolysis and PARP-dependent repair with it.

Translatability. Moderate: eleven chondrosarcoma lines with low-nanomolar IC50s in nearly half, and NAPRT methylation rises with grade, but no chondrosarcoma animal work exists.

Developability. Middling. FK866 and GMX1778 failed phase 1 on thrombocytopenia and retinal toxicity;

KPT-9274 and OT-82 are newer, and no clinical NAPRT assay exists.

Counter-productive MoA. Moderate (Dietary or supplemental nicotinic acid reopens the salvage route and would blunt the whole effect.)

Key risks.

- The clinical record of NAMPT inhibitors is poor: dose-limiting thrombocytopenia, retinal toxicity, and no efficacy signal in unselected phase 1 populations.
- Dietary or supplemental nicotinic acid rescues NAPRT-competent cells and would blunt the effect in anyone not on controlled intake.
- No in-vivo chondrosarcoma data at all; the whole chondrosarcoma case is cell-line work from one group.
- NAPRT promoter methylation has never been run prospectively as a selection biomarker, and no clinical-grade assay exists.
- Sensitivity was measured in conventional lines, so whether a dedifferentiated component keeps the same NAPRT silencing is unknown.

Open questions.

- Does a dedifferentiated component retain the NAPRT silencing that was measured in conventional lines?
- Can NAPRT methylation be assayed on routine clinical material, which is the prerequisite for selecting anyone?

Key references. pmid:28860121; pmid:26678339; pmid:32606138

4. Class I HDAC inhibitor priming (vorinostat, romidepsin, entinostat) before BH3-mimetic or glutaminolysis blockade (combination)

Target: High-grade chondrosarcoma with dedifferentiated features: epigenetic control of the apoptotic and metabolic set-point **Stage:** In vitro **Evidence strength:** Emerging

The value sits entirely in what it enables, and the agent it would enable shares its dose-limiting thrombocytopenia.

Venneker screened epigenetic compounds across five chondrosarcoma lines and HDAC inhibition was the standout, in both 2D and 3D and independent of IDH mutation status or subtype. Independence from IDH status matters here, because her IDH result does not exist yet. The more useful half of the paper is the second result: class I HDAC inhibition made the cells sensitive to BCL-2-family and glutaminolysis inhibitors they otherwise tolerated. That puts this underneath the BCL-xL idea at rank 1 rather than beside it. Two things hold it here. On its own it is unlikely to do much, since HDAC inhibitors have failed repeatedly as single agents in solid tumours including sarcoma, and the combination it enables shares its dose-limiting toxicity. Thrombocytopenia from a class I HDAC inhibitor and thrombocytopenia from a BCL-xL agent are the same problem arriving twice. All the chondrosarcoma evidence is cell culture, with no animal work and no dedifferentiated models, and priming schedules are notoriously sensitive to sequence and interval, none of which has been worked out.

Mechanism. Class I HDACs hold down transcription of BH3-only proteins and shape the metabolic programme these cells run. Inhibiting them moves the apoptotic set-point, so a BCL-2-family or glutaminase inhibitor arriving afterwards meets a cell already closer to the death threshold.

Translatability. Moderate: five chondrosarcoma lines in 2D and 3D, with sensitisation independent of IDH status

or subtype, but nothing has left cell culture.

Developability. Near on chemistry: vorinostat and romidepsin are approved in cutaneous T-cell lymphoma. The obstacle is a combination phase 1, not a molecule.

Counter-productive MoA. Moderate (Shares dose-limiting thrombocytopenia with the BCL-xL partner it is meant to enable, so both may end up underdosed.)

Key risks.

- The partner agent and the primer share thrombocytopenia as their dose-limiting toxicity, which is a real problem for this exact pairing.
- HDAC inhibitors have failed repeatedly as single agents in solid tumours including sarcoma.
- All chondrosarcoma evidence is cell culture: no animal work, no dedifferentiated models.
- Priming schedules depend heavily on sequence and interval, and neither has been worked out here.

Open questions.

- Is there a schedule that separates the two thrombocytopenia signals enough to dose both drugs properly?
- Does the priming effect hold in a dedifferentiated model, or only in the conventional lines screened?

Key references. pmid:33266275; pmid:29100385; pmid:17200366

5. Nitrogen-containing bisphosphonates (zoledronic acid, minodronate) as antitumour agents in the bone niche (repurposed drug)

Target: Large acetabular lesion with pathologic fracture: the osteolytic bone niche **Stage:** In vivo (animal)

Evidence strength: Emerging

The only in-vivo evidence that anything slows chondrosarcoma at a bone site, resting on a 2006 rat study nobody replicated and no trial followed.

One caveat belongs at the front of this row: bisphosphonate or denosumab use for skeletal-related events is separately standard supportive care and belongs to the standard-of-care track. This row is only about the antitumour hypothesis, which is a different and much thinner claim. The Swarm rat work is the only in-vivo evidence that anything slows chondrosarcoma at a bone site. Zoledronic acid slowed primary tumour development, slowed recurrence after intralesional curettage, and improved overall survival, and that pairing of primary growth with post-curettage recurrence maps unusually well onto a large acetabular lesion where local control is contested. Lai adds a tumour-cell-directed anti-invasive effect rather than a purely osteoclast-mediated one. The read-through is what holds it down. Zoledronic acid added to chemotherapy failed to improve outcomes in the OS2006 osteosarcoma trial, the closest clinical comparison available, and the rat result is from 2006 with no replication and no follow-up. Osteonecrosis of the jaw and atypical femoral fracture also matter more, not less, in someone who already has a pathologic fracture of a weight-bearing joint.

Mechanism. Nitrogen-containing bisphosphonates block farnesyl pyrophosphate synthase, disable osteoclasts, and interrupt the resorption-driven release of bone-matrix growth factors that feeds a tumour sitting in bone. A direct anti-invasive effect on chondrosarcoma cells, with reduced MMP-2 secretion, has also been reported.

Translatability. Moderate: an orthotopic rat model with primary-growth and post-curettage endpoints, which maps

onto a contested acetabular lesion better than most models do.

Developability. Nearest of any row: generic and already used in oncology. The hard part is separating an antitumour effect from the skeletal-event benefit.

Counter-productive MoA. Moderate (Osteonecrosis of the jaw and atypical fracture risk land on someone who already has a pathologic fracture.)

Key risks.

- Zoledronic acid added to chemotherapy failed to improve outcomes in the OS2006 osteosarcoma trial, the closest clinical read-through available.
- Osteonecrosis of the jaw and atypical femoral fracture matter more, not less, in a patient with an existing pathologic fracture of a weight-bearing joint.
- The efficacy evidence is a single rat study from 2006 with no replication.
- Because these drugs are already standard supportive care for skeletal events, any antitumour claim is hard to isolate and this row should not be read as duplicating that indication.

Open questions.

- Could any trial design separate an antitumour effect from the skeletal-related-event benefit these drugs already deliver?
- Does the Swarm rat result reproduce in a dedifferentiated model, or in any model built since 2006?

Key references. pmid:16570273; pmid:17640487

6. 5'-methylschweinfurthin G, an academic natural-product analogue active in matrix-rich chondrosarcoma (small molecule)

Target: Cartilaginous matrix and drug penetration in a bulky high-grade lesion **Stage:** In vivo (animal) **Evidence strength:** Emerging

The largest in-vivo effect size in this scan rests on a single 2018 rat study, with no replication, no follow-up, and no drug-development package.

This compound was built against the exact problem a bulky cartilaginous mass poses, a collagen II and proteoglycan-rich tumour that resists drug penetration. In the Swarm rat, which carries that matrix phenotype, repeated dosing cut tumour mass by 83% over 17 days and eliminated tumour outright in three of ten animals, with reduced proliferation and increased cell death on histology. The selectivity number is the more interesting one: an IC50 of 35 nanomolar against tumour cells against 11 micromolar for freshly isolated chondrocytes. That is real in-vivo activity in the right matrix context, and very little else in this corner of the field has produced any. Everything else about it is thin. One paper, one group, 2018, no independent replication and no visible follow-up. The model is a rat tumour with a well-differentiated cartilage phenotype, which is close to the matrix problem but a long way from a dedifferentiated human tumour. The mechanism is inferred from the schweinfurthin class rather than demonstrated here, and systemic disruption of cholesterol trafficking has unexplored toxicity. I listed it for the in-vivo result, which is real. Nothing about the development path is.

Mechanism. Schweinfurthins act on oxysterol-binding-protein-family lipid transfer proteins and disturb cholesterol and sphingolipid trafficking, a dependency some tumour lineages carry and normal cells largely do not.

Translatability. Low: real in-vivo activity at 83% mass reduction, but in a rat tumour with a well-differentiated cartilage phenotype, not a dedifferentiated human one.

Developability. Very far: no IND, no pharmacokinetics, no formal toxicology, and demanding synthesis. The whole preclinical package would have to be built first.

Counter-productive MoA. Low (Cholesterol and sphingolipid trafficking disruption is systemic, with no toxicology to say where the therapeutic window sits.)

Key risks.

- One paper from one group in 2018, with no independent replication and no visible follow-up.
- A rat chondrosarcoma with a well-differentiated cartilage phenotype is close to the matrix problem and far from her tumour.
- The lipid-transfer mechanism is inferred from the compound class rather than demonstrated in this work.
- Systemic disruption of cholesterol trafficking has unexplored toxicity and no therapeutic window.

Open questions.

- Does the selectivity window hold against a dedifferentiated human tumour rather than a well-differentiated rat one?
- Has anyone tried to replicate the 83% result in the eight years since it was published?

Key references. pmid:28960476

7. Anti-miR-181a oligonucleotide delivered on a nanopiece platform (oligonucleotide)

Target: Multiple dispersed pulmonary metastases: the angiogenic and invasive programme behind them **Stage:** In vivo (animal) **Evidence strength:** Emerging

Aimed squarely at the part of her disease local therapy cannot reach, on the strength of a 32% volume reduction from one laboratory.

Her pulmonary nodules are bilateral, multiple and dispersed, which closes off metastasectomy and stereotactic radiotherapy and makes systemic control the operative question for the lungs. That is the gap this row addresses. Sun delivered anti-miR-181a systemically in chondrosarcoma xenografts and got a 32% reduction in tumour volume by day 38, with survival rising from 23% to 45% and the expected molecular readouts on RGS16, VEGF and MMP1. It is a different modality aimed at a metastasis-directed mechanism, which is why it is on the page at all. The effect size is modest enough that it would not justify a trial on its own, the work is one laboratory with no independent replication, and subcutaneous xenografts say very little about whether an agent reaches established pulmonary metastases. miR-181a expression in her tumour is unknown and there is no clinical assay for it. The oligonucleotide chemistry is routine; the nanopiece carrier is the part that would take years.

Mechanism. miR-181a is overexpressed in high-grade chondrosarcoma and represses RGS16, which normally dampens CXCR4 signalling. Sequester the miRNA and RGS16 comes back, CXCR4 quiets down, and MMP1 and VEGF output falls. The target is angiogenesis and invasion rather than proliferation.

Translatability. Low: human chondrosarcoma xenografts with systemic dosing and the expected molecular

readouts, but subcutaneous tumours say little about established lung metastases.

Developability. Far: the oligonucleotide chemistry is routine, the nanopiece carrier is not. No IND, no toxicology, no regulatory history for the platform.

Counter-productive MoA. Low (Antiangiogenic and anti-invasive rather than cytotoxic, so burden may be contained without being reduced.)

Key risks.

- A 32% volume reduction is modest and would not on its own justify a trial.
- Single laboratory, no independent replication.
- miR-181a expression in her tumour is unknown, and there is no clinical assay for it.
- Subcutaneous xenografts say little about whether the agent reaches established pulmonary metastases.

Open questions.

- Does anti-miR-181a do anything to established lung metastases, as opposed to subcutaneous tumours?
- Is there any route to a clinical miR-181a assay that would let a patient be selected?

Key references. pmid:31341031

8. Plectin inhibition with Plecstatin-1 in dedifferentiated chondrosarcoma (small molecule)

Target: Dedifferentiated component: cytolinker-dependent migration and EMT **Stage:** In vitro **Evidence strength:** Speculative

One preprint, one cell line, and readouts that stop at migration. The model it came from may matter more to this field than the compound does.

Almost nothing in this literature is generated in a genuinely dedifferentiated model, and that scarcity is the whole argument for this row. PUMC-DDCS1 is a newly established patient-derived dedifferentiated line with matched whole-genome, exome, bisulfite and transcriptome profiling, full tumorigenicity in immunodeficient mice, and sensitivity to cisplatin and doxorubicin, which is a useful internal control because it reproduces the chemosensitivity that makes this histology different. Within that model, plectin inhibition was the one non-cytotoxic hit. It is the only compound nomination in this scan that came out of a dedifferentiated chondrosarcoma model rather than being imported into one. That is where the case ends. A single unreplicated preprint, one cell line, and readouts that stop at migration, invasion and EMT markers rather than survival or tumour burden. There is no clinical-grade plectin inhibitor, no toxicology package, and ruthenium organometallics bring their own formulation problems. Plectin is broadly expressed in skin, muscle and epithelium, so on-target toxicity is a live question with no data behind it. The cell line may end up worth more to the field than the compound is.

Mechanism. Plectin cross-links intermediate filaments to actin and to focal adhesions, a common dependency in migratory mesenchymal-state tumour cells. Plecstatin-1, a ruthenium organometallic, binds plectin and pulls that scaffold apart; in the reported line the downstream effect ran through AKT and ERK signalling.

Translatability. Low but uniquely well matched: the only nomination generated inside a patient-derived dedifferentiated chondrosarcoma line rather than imported into one.

Developability. Very far: an academic ruthenium tool compound with no IND, no toxicology, and formulation problems particular to organometallics.

Counter-productive MoA. Low (Migration and invasion endpoints only, so the compound may reshape behaviour without reducing tumour burden.)

Key risks.

- Single unreplicated preprint, not peer reviewed.
- One cell line, and the anti-plectin effect was measured on migration and invasion rather than survival or tumour burden.
- Plectin is broadly expressed in skin, muscle and epithelium, so on-target toxicity is a live question with no data.
- No in-vivo efficacy work at all.

Open questions.

- Does plectin inhibition change tumour burden in this model, or only cell behaviour in a dish?
- Will PUMC-DDCS1 be shared widely enough to support the functional work the field is missing?

Key references. doi:10.21203/rs.3.rs-10674626/v1

9. Biomarker-selected SMO inhibition (vismodegib) in PTCH1-altered chondrosarcoma, nominated by patient-derived organoids (repurposed drug)

Target: Unmeasured molecular landscape: PTCH1 loss-of-function as a selection marker **Stage:** In vitro
Evidence strength: Speculative

Genotype selection is the one thing the failed unselected chondrosarcoma trial never tried, but PTCH1 loss is not a recognised recurrent event in this disease.

Hedgehog inhibition has a bad name in chondrosarcoma and earned it, in an unselected phase 2 (NCT01310816) that went nowhere. What makes the Takami preprint worth a line is that it changes the selection question rather than the drug. The first patient-derived chondrosarcoma organoids, validated against their parental tumours histologically and genetically, carried PTCH1 loss-of-function and responded strongly to vismodegib in culture. Since she has had no genomic profiling at all, PTCH1 would come along free on the comprehensive panel the case already needs for the IDH question, which costs nothing extra to ask. The reasons to be sceptical are substantial. Two organoid lines in an unreviewed preprint cannot establish that PTCH1 alterations occur at usable frequency, and PTCH1 loss-of-function is not a recognised recurrent event in chondrosarcoma genomic series, so the prior is low. The drug activity was in vitro only, with no in-vivo treatment reported, and neither organoid is described as dedifferentiated. SMO inhibitor toxicity is also poorly tolerated over months, which matters in a patient whose performance status was assumed rather than measured and may be worse than recorded.

Mechanism. PTCH1 restrains SMO. Loss-of-function releases that restraint and hedgehog signalling runs constitutively. Vismodegib binds SMO and shuts the pathway down, which only helps where the lesion sits upstream of SMO.

Translatability. Low: two patient-derived organoid lines in a preprint, vismodegib activity in culture only, and

neither organoid described as dedifferentiated.

Developability. Near on molecule, weak on premise: vismodegib is approved, and PTCH1 rides along free on the profiling panel this case already needs.

Counter-productive MoA. Moderate (SMO inhibitor toxicity is hard to sustain for months at an already-limited performance status.)

Key risks.

- Unselected hedgehog inhibition already failed in advanced chondrosarcoma, so the burden of proof on any revival is high.
- Two organoid lines, preprint, no peer review, and no in-vivo drug treatment reported.
- PTCH1 loss-of-function is not a recognised recurrent event in chondrosarcoma genomic series, so the prior is low before any test is run.
- The organoids are not described as dedifferentiated, and the activity was in culture only.
- SMO inhibitor toxicity is hard to sustain over months, and her performance status was assumed rather than measured.

Open questions.

- At what frequency, if any, do PTCH1 alterations actually occur across chondrosarcoma genomic series?
- Does vismodegib do anything in vivo in a PTCH1-altered chondrosarcoma model?

Key references. doi:10.21203/rs.3.rs-7826834/v1

10. Blocking the interstitial-shear to MMP-7/MMP-12 axis (IGF-2, VEGF-B/D, IL-1beta or the proteases as nodes) (conceptual strategy)

Target: Large acetabular lesion with pathologic fracture: the mechanical environment as a driver, not only a complication **Stage:** In vivo (animal) **Evidence strength:** Emerging

Target biology worth measuring in her tumour rather than an intervention anyone can offer, and the loading data cut in both directions.

Two features dominate this case, a load-bearing pelvic lesion that has already fractured and disseminated pulmonary nodules, and this is the one line of work that connects them mechanistically. Wang and Guan built the mechanical arm in human chondrosarcoma lines and showed the shear-induced protease programme is sufficient to promote lung colonisation in vivo. Fan supplies the uncomfortable half of the picture from a breast-cancer osteolysis model: loading is not monotonically good or bad, with low-intensity load suppressing osteolysis and higher intensity reversing the benefit. That is as much a caution about weight-bearing and rehabilitation decisions as it is a drug target. There is, however, no drug. Broad-spectrum MMP inhibitors were abandoned decades ago for musculoskeletal toxicity without efficacy, nobody has built a selective MMP-7 or MMP-12 agent for oncology, and the upstream nodes that do have agents have never been tested against this readout. The link from her fracture and altered loading to the shear stresses used in a bioreactor is an inference, not a measurement. Read this as target biology worth measuring rather than as anything anyone can offer.

Mechanism. Interstitial fluid flow across chondrosarcoma cells raises cAMP and IL-1beta and induces IGF-2 and VEGF-B/D, which transactivate MMP-7 and MMP-12 through PI3K, ERK, p38 and JNK. The resulting proteases

drive motility, matrix invasion and, in the mouse, lung colonisation.

Translatability. Low: human chondrosarcoma lines under controlled shear with lung colonisation in vivo, but the link to her fractured, loaded acetabulum is inference.

Developability. Nothing to develop: no selective MMP-7 or MMP-12 agent exists, and broad-spectrum inhibitors were abandoned decades ago for musculoskeletal toxicity.

Counter-productive MoA. High (Blocking matrix proteolysis could tighten a matrix that already impedes drug delivery, cutting against the penetration ideas elsewhere here.)

Key risks.

- The MMP-inhibitor field has a bad history, and a selective agent would be starting from zero.
- The link from her pathologic fracture and altered load to bioreactor shear stresses is an inference, not a measurement.
- The loading-intensity data come from a breast-cancer osteolysis model, not a cartilage tumour.
- Blocking matrix proteolysis could plausibly worsen drug penetration in a matrix-dense tumour, which cuts against the delivery ideas elsewhere in this scan.

Open questions.

- Are MMP-7 and MMP-12 actually elevated in a fractured, loaded human chondrosarcoma, or only in a bioreactor?
- Does the loading-intensity finding hold in a cartilage tumour, given what it would imply for rehabilitation?

Key references. pmid:25435370; pmid:25823818; pmid:32128277

11. Blocking the collagen-LAIR-1 inhibitory axis in a collagen II-rich tumour (LAIR-2 Fc or equivalent) (conceptual strategy)

Target: Dedifferentiated chondrosarcoma immune microenvironment: collagen as an active immunosuppressive ligand **Stage:** Conceptual **Evidence strength:** Speculative

A hypothesis about why checkpoint blockade disappoints in this disease, with no cartilage-tumour data behind it and a direction of effect that could invert.

The immunology in this case points at collagen and myeloid infiltration as the operative resistance mechanisms rather than T-cell exhaustion, and LAIR-1 is the one receptor that ties those two observations into a single mechanism. Lebbink mapped multiple high-affinity LAIR binding sites specifically on collagens II and III, and collagen II is the defining matrix protein of a cartilage tumour. Peng separately showed collagen driving anti-PD-1 resistance through LAIR1-dependent CD8 exhaustion, which would explain checkpoint failure in this disease better than PD-L1 levels do. Being blunt about what this is: nobody has looked at LAIR-1 in a cartilage tumour, so it is a hypothesis about why the existing immunotherapy story disappoints, not a treatment. A clinical-grade LAIR-2 Fc fusion exists and has been dosed in other solid tumours, so the molecule would not need inventing, but its activity elsewhere has been modest. There is also a real risk the direction of effect is simply wrong. Releasing LAIR-1 inhibition on monocytes could amplify a tumour-supportive myeloid programme as easily as an antitumour one, and in syngeneic chondrosarcoma models it is macrophage depletion that helps. Collagen II in chondrosarcoma is often abnormal, and whether it keeps the LAIR-binding motifs is unknown. This belongs on a correlative-science

list before it belongs on a therapy list.

Mechanism. LAIR-1 is an ITIM-bearing inhibitory receptor on T cells, NK cells, monocytes and dendritic cells, and its physiological ligand is collagen. Engagement suppresses effector function. A soluble LAIR-2 decoy or a blocking antibody competes collagen away from the receptor and lifts that brake.

Translatability. Low, and zero in cartilage: collagen II is a high-affinity LAIR-1 ligand biochemically, but nobody has stained a chondrosarcoma for the receptor.

Developability. Molecule exists, target does not: a LAIR-2 Fc fusion has been dosed in other solid tumours, with modest activity so far.

Counter-productive MoA. High (Lifting LAIR-1 inhibition on monocytes could amplify a tumour-supportive myeloid programme, the opposite of the intended effect.)

Key risks.

- Zero data in cartilage tumours; the mechanistic fit is inferred from collagen II biochemistry.
- Releasing LAIR-1 inhibition on monocytes could amplify a tumour-supportive myeloid programme, and syngeneic chondrosarcoma data show macrophage depletion helping rather than macrophage activation.
- Collagen II in chondrosarcoma is often abnormal, and whether it retains the LAIR-binding motifs is unknown.
- The clinical-grade agent's activity in other tumour types has been modest.

Open questions.

- Do chondrosarcoma tumour-infiltrating myeloid and T cells express LAIR-1 at all, which nobody has looked at?
- Does the abnormal collagen II these tumours make still carry the LAIR-binding motifs?

Key references. pmid:19345263; pmid:32908154; pmid:21955987; pmid:34121658

12. Targeting the SOX9-to-RUNX2 differentiation switch that defines the dedifferentiated component (conceptual strategy)

Target: The dedifferentiation transition itself **Stage:** In vitro **Evidence strength:** Speculative

Named to mark the gap precisely: the transition that defines her tumour is real and measurable, and currently undruggable in either direction.

This is the honest answer to whether anything nominates a target inside the dedifferentiation transition, and the answer is almost nothing. What exists: Tang's observation that SOX9 and RUNX2 expression separate conventional from dedifferentiated lines, and the functional result that knocking SOX9 out of a chondrosarcoma line changes ploidy and shifts sensitivity to standard treatment modalities, so differentiation state is not merely descriptive and does move drug response. Nicole's multi-omic classification then puts the balance between differentiation and cell-cycle activation at the centre of chondrosarcoma subtyping. Together that is enough to say the axis is real, and enough to say nobody has drugged it. The direction of benefit is also unclear. Forcing redifferentiation might restore the chemoresistant cartilage phenotype rather than help, which would be the opposite of what her histology's chemosensitivity offers, and the one knockout experiment produced polyploidy, which is not obviously a therapeutic outcome. Transcription factors of this class have no tractable chemistry, all the human evidence is correlative, and the field has a handful of dedifferentiated lines and no genetically engineered

mouse. I include it to name the gap precisely, not to propose a treatment.

Mechanism. SOX9 maintains the chondrocytic programme and RUNX2 drives the osteoblastic and mesenchymal one, and dedifferentiated lines carry the opposite balance to conventional ones. The strategy would be to re-impose the chondrocytic programme on the dedifferentiated clone, or to attack a dependency only the RUNX2-high state has. Neither currently has a molecule.

Translatability. Low: correlative human expression data plus one knockout that changed ploidy. No compound exists to test the axis in either direction.

Developability. Nothing to develop: SOX9 and RUNX2 have no clinical-grade ligands and no degrader programmes, and the field has only a handful of models.

Counter-productive MoA. High (Forcing redifferentiation could restore the chemoresistant cartilage phenotype, undoing the chemosensitivity this histology actually offers.)

Key risks.

- Direction of benefit is unclear: forcing redifferentiation might restore the chemoresistant cartilage phenotype rather than help.
- The one knockout experiment produced polyploidy, which is not obviously a therapeutic outcome.
- Transcription factors of this class have no tractable chemistry.
- All of the human evidence is correlative.

Open questions.

- Would re-imposing the chondrocytic programme help or simply hand the tumour its chemoresistance back?
- Can a functional model of the transition be built at all, given how few dedifferentiated lines exist?

Key references. pmid:33076370; pmid:20683191; pmid:31604924; pmid:22928481

13. Losartan or other antifibrotic matrix normalisation to improve drug delivery into cartilaginous tumour (repurposed drug)

Target: Matrix-mediated chemoresistance in a bulky cartilaginous mass **Stage:** Conceptual **Evidence strength:** Speculative

Included as the honest answer to a direct question about matrix normalisation: the mechanism does not match the matrix in question.

Matrix-mediated chemoresistance is a reasonable thing to ask about in a bulky cartilaginous mass, and this row is the answer rather than a proposal. Nothing in cartilage tumours has moved past the hyaluronidase failure. Losartan is the strongest repurposing candidate in the general matrix-normalisation literature, with reduced collagen, improved perfusion and better chemotherapy efficacy in ovarian and pancreatic models. The mismatch is the problem. That work targets fibroblast-derived collagen I and hyaluronan, while the chondrosarcoma barrier is tumour-cell-made collagen II and aggrecan laid down by the malignant chondrocytes themselves, a different problem with a different enzyme set. No matrix-normalisation agent has ever been tested in a cartilage tumour, in any model or any trial. The PEGPH20 experience is a standing warning that improving delivery on paper does not improve outcomes and can worsen them, and loosening a matrix that is physically containing tumour has an

obvious theoretical downside nobody has excluded. Hypotension is also not a trivial side effect in someone with a painful pathologic fracture and limited mobility. It earns its place as the honest answer to a direct question, not because anyone could test it here.

Mechanism. Angiotensin receptor blockade suppresses TGF-beta and CCN2 signalling in stromal fibroblasts, reduces collagen I and hyaluronan deposition, lowers solid stress, decompresses vessels, and lets more chemotherapy reach the tumour interior. The logic is delivery, not cytotoxicity.

Translatability. Low, cross-tumour only: the ovarian and pancreatic work targets fibroblast-derived collagen I, while the cartilage barrier is tumour-made collagen II and aggrecan.

Developability. Molecule is generic and oral; the missing piece is a chondrosarcoma model with a measurable penetration readout, which nobody has built.

Counter-productive MoA. High (Loosening a matrix that physically contains tumour is an unexcluded downside, and PEGPH20 showed delivery gains can worsen outcomes.)

Key risks.

- Mechanism mismatch: cartilage matrix is collagen II and aggrecan, not the fibroblast-derived collagen I losartan acts on.
- The PEGPH20 experience warns that improving delivery on paper does not improve outcomes and can worsen them.
- Hypotension is not a trivial side effect in someone with a painful pathologic fracture and limited mobility.
- Loosening a matrix that is physically containing tumour has an obvious theoretical downside nobody has excluded.

Open questions.

- Is there any antifibrotic that acts on tumour-cell-made collagen II and aggrecan rather than fibroblast collagen I?
- Could a 3D pellet system give a penetration readout worth measuring before anyone dosed a patient?

Key references. pmid:30659155